

# A method that works for every secret

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

## 1. Explore three places.

Play a round without a test limit. Keep the old tests and scores visible. Then try this investigation.

## 2. Change exactly one place.

Compare RRR with BRR. Can the score stay the same? What does an increase or a decrease tell you?

| Test | My code | Score | Still possible |
|------|---------|-------|----------------|
| 1    |         |       |                |
| 2    |         |       |                |
| 3    |         |       |                |
| 4    |         |       |                |

## 3. Three tests, then name the secret.

Try RRR, BRR, and RBR. How can these reveal all three places? Check your method on a partner’s secret.

### Why my method always determines the third place too:

**Counting rule:** Naming a known secret costs no test here. In the original game, submitting it as a winning guess may cost one more turn.

Build first. Explain by pointing, talking, or drawing.

# Can anyone always do it in two?

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

## 4. The difficult first answer.

Suppose RRR scores 2. Exactly these secrets remain:

BRR      RBR      RRB

## 5. Can a last test separate all three?

Try rows of the table and circle repeated scores. What is the only feature that changes among the three secrets? Explain why a test can tell only whether the secret's B is in one of its B positions. **Stop calculating rows once you can explain why at most two scores occur.**

| Test | BRR | RBR | RRB |
|------|-----|-----|-----|
| RRR  |     |     |     |
| RRB  |     |     |     |
| RBR  |     |     |     |
| RBB  |     |     |     |
| BRR  |     |     |     |
| BRB  |     |     |     |
| BBR  |     |     |     |
| BBB  |     |     |     |

## 6. Does a different first test help?

Try BRB. In *both* a secret and every test, swap R/B at places 1 and 3. The first test becomes RRR. Check that each matching place still matches. Do this at every B position of any first test. Why are all scores unchanged? Discuss with an adult.

**Why no first test can guarantee success in only two tests:**

**What we proved:** Task 5 rules out two tests starting with RRR. With task 6, it covers every first test; together with page 1, three is optimal.

Build first. Explain by pointing, talking, or drawing.